

CALIFORNIA STATE UNIVERSITY, NORTHRIDGE

ACTIVE PERCEPTION FOR BRAIN TUMOR DETECTION
USING HYBRID SELF-SUPERVISED AND REINFORCEMENT LEARNING

A thesis submitted in partial fulfillment of the requirements for the degree of Master of Science
in Computer Science

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TABLE OF CONTENTS

SIGNATURE PAGE	ii
ACKNOWLEDGMENTS	iii
LIST OF TABLES	vi
LIST OF FIGURES.....	vii
CHAPTER 1: INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	3
1.3 Research Contributions	4
1.4 Thesis Organization	5
CHAPTER 2: RELATED WORK	6
2.1 Chapter Overview.....	6
2.2 Supervised Brain Tumor Segmentation.....	6
2.2.1 Encoder-Decoder Architectures	6
2.2.2 Attention and Multi-Scale Methods.....	7
2.2.3 Transformer-Based Segmentation	7
2.3 Self-Supervised Representation Learning.....	8
2.3.1 Contrastive Learning Frameworks	8
2.3.2 SSL in Medical Imaging	9
2.4 Reinforcement Learning for Visual Decision-Making	9
2.4.1 Deep Reinforcement Learning Foundations	9
2.4.2 Reinforcement Learning for Medical Image Analysis	10
2.5 Why Reinforcement Learning Over Alternative Paradigms	10
2.6 Summary.....	12
CHAPTER 3: DATASET AND EXPERIMENTAL SETUP	13
3.1 The BraTS 2020 Dataset	13
3.2 Why 2D Axial Slices Rather Than 3D Volumes	14
3.3 Preprocessing Pipeline	15
3.4 Data Split Strategy	16
3.5 Evaluation Protocol	17
CHAPTER 4: PROPOSED METHODOLOGY.....	18
4.1 Framework Overview and Novelty	18
4.2 Why Reinforcement Learning Outperforms Alternative Paradigms.....	19
4.3 Theoretical Foundations: Active Perception as POMDP.....	20
4.3.1 The Limitation of Static Inference	20
4.3.2 Partially Observable Markov Decision Process Formulation	20

4.4 Self-Supervised Representation Learning: SimCLR	21
4.4.1 Contrastive Learning as Mutual Information Maximization	21
4.4.2 Architecture: Why ResNet-18	22
4.4.3 Projection Head and Embedding Dimensions	22
4.4.4 Augmentation Strategy	23
4.5 Multi-Scale State Representation	24
4.5.1 Information-Theoretic Justification	24
4.5.2 Clinical Analogy	25
4.6 Reinforcement Learning Foundations	25
4.6.1 Markov Decision Process Formulation	25
4.6.2 Double DQN: Correcting Overestimation Bias	26
4.6.3 Dueling Architecture	26
4.7 Reward Function Design and Clinical Calibration	27
4.7.1 Reward Structure and Asymmetric Penalties	27
4.7.2 FN Penalty Calibration Through Sensitivity Analysis	28
4.7.3 Step Penalty Rationale	29
4.8 The RL Environment: SliceEnv	29
4.8.1 State Space	29
4.8.2 Action Space and Zoom Mechanics	30
4.9 Dueling Double DQN Architecture and Training	31
4.9.1 Network Architecture	31
4.9.2 Training Stability Techniques	32
CHAPTER 5: EXPERIMENTAL RESULTS	33
5.1 SimCLR Pre-training Analysis	33
5.3 Final Performance Metrics	37
5.4 Zoom Trajectory Visualization	39
5.6 Comprehensive Model Comparison	43
CHAPTER 6: DISCUSSION	45
6.1 Interpretation of Results	45
6.2 The Multi-Scale State Contribution	46
6.3 Reward Design and Clinical Calibration	46
6.4 Comparison to Prior Work	47
6.5 Future Work	47
CHAPTER 7: CONCLUSION	49
REFERENCES	51

LIST OF TABLES

Table 1: System Implementation Overview	23
Table 2: SimCLR Pre-training Configuration	31
Table 3: Reward Function Design and Justification	35
Table 4: RL Environment (SliceEnv) Specification	38
Table 5: RL Training Hyperparameters	39
Table 6: Validation Metrics at Key Training Checkpoints (Seed 42)	44
Table 7: Final Evaluation Metrics - APBTD System (Mean, 3 Seeds)	45
Table 8: Ablation: Zoom vs. No-Zoom (Mean, 3 Seeds)	49

LIST OF FIGURES

Figure 1: Sample BraTS 2020 axial MRI slices (tumor and normal)	21
Figure 2: SimCLR contrastive pre-training loss (NT-Xent) over epochs	21
Figure 3: SSL+RL training convergence over 120,000 frames across three seeds	27
Figure 4: System architecture of the APBTD framework	36
Figure 5: ROC curves - CNN Baseline, SSL+RL Zoom, SSL+RL No-Zoom	41
Figure 6: Precision-Recall curves - CNN Baseline, SSL+RL Zoom, No-Zoom	43
Figure 7: Comprehensive model comparison across five evaluation metrics	44
Figure 8: Grad-CAM activation maps for CNN Baseline on tumor slices	45
Figure 9: SSL SimCLR feature similarity maps for BraTS 2020 tumor slices	46
Figure 10: RL agent zoom trajectory for a tumor-positive slice	46
Figure 11: RL agent zoom trajectory for a normal slice	47
Figure 12: Ablation results: Zoom Enabled vs. No-Zoom	47
Figure 13: FN penalty sensitivity analysis (sensitivity/specificity trade-off)	48
Figure 14: Confusion matrices for three model configurations	49
Figure 15: Exploratory data analysis of the BraTS 2020 dataset	50

Abstract

Active Perception for Brain Tumor Detection Using Hybrid Self-Supervised and Reinforcement Learning

By

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Master of Science in Computer Science

Accurate detection of brain tumors from Magnetic Resonance Imaging (MRI) is a critical task in computational medical imaging. Though supervised deep learning approaches have proven to perform well, they often demand a large amount of expertly labeled training data, which is both costly and challenging to obtain in practice. In this thesis, we propose the Active Perception for Brain Tumor Detection (APBTD) framework that alleviates both the labeling scarcity constraint and the need to localize a region of interest in MRI data using a principled approach to extracting and adapting a representation learned via self-supervised learning, as part of an active and efficient search into the unknown environment.

In Stage 1 of the pre-learning frontier of the Active Perception for Brain Tumor Detection framework, a ResNet-18 encoder is trained on as many as 38,014 unlabeled axial MRI slices randomly sampled from the BraTS 2020 multi-modal MRI datasets, using the SimCLR contrastive learning algorithm. It augments each slice twice to form a positive pair and trains the encoder to pull representations of the same image closer together, with the NT-Xent loss. The frozen encoder maps the slice into a 512-dimensional encoding of structural and intensity-based radiometric

appearance features without annotation. A global-local multi-scale state vector is constructed by appending the 512-dimensional encoding of the whole image to the 512-dimensional encoding of the current zoomed-in region to produce a 1024-dimensional state vector for the reinforcement learning agent. In the second stage, a Dueling Double Deep Q-Network (Dueling Double DQN) agent operates within a custom SliceEnv environment, where it may take up to five sequential spatial zoom actions, top-left, top-right, bottom-left, and bottom-right quadrant crops, before committing to a binary tumor-or-normal prediction. This formulation casts the diagnostic process as a Partially Observable Markov Decision Process, mirroring the coarse-to-fine inspection strategy that expert radiologists employ when reviewing MRI volumes. A carefully designed asymmetric reward function, with penalties calibrated through empirical sensitivity analysis, balances detection sensitivity against false positive rate in a manner appropriate for a clinical screening context.

Experiments consist of 120,000 training frames, and the experiment was repeated three times by varying the random seed for each run to evaluate the stability of our results. The Active Perception for Brain Tumor Detection system clocks an average accuracy of 84.6%, sensitivity of 80.1%, specificity of 83.5%, F1 of 80.1%, and a representative ROC-AUC of 0.918. This is a considerable advance over an SSL-only system tested with a linear probe (77.7% accuracy, 85.5% F1), showing that the exploration part of the system introduced by reinforcement learning provides a strong discriminative signal. Ablation studies also confirm that the zoom mechanism alone adds +1.58% to accuracy, +4.35% to sensitivity, and +1.66% to F1 relative to a variant without zooming. However, examination of the proposed zooming trajectories demonstrates qualitatively that the agent does indeed zoom into diagnostic regions of the image. The approach does not rely on pixel-level annotations and is "low-cost" for low-budget or triage-based applications.

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Brain tumors are one of the most morbid categories of central nervous system disease. Brain tumors are graded by the World Health Organization according to a four-tier system as benign (Grade I) or malignant (Grades II-IV), with the most aggressive (and common) form being glioblastoma multiforme (GBM, Grade IV). Despite intensive multimodal treatment, the median survival of GBM remains less than 15 months, highlighting the need for early detection and proper characterization [1-5].

Magnetic Resonance Imaging (MRI) is the preferred imaging method for the evaluation of brain tumors because of its unrivalled soft tissue contrast resolution, non-ionizing nature, and ability to acquire multiple intrinsic tissue properties in a single scan. On a minimum, current clinical MRI image acquisition protocols used to evaluate gliomas include a series of four images: T1-weighted images for anatomical delineation, T1 contrast-enhanced (T1ce) imaging to depict blood-brain barrier breakdown and a surrounding rim of infiltration, T2-weighted images to show enhancing regions of elevated signal intensity as well as peritumoral vasogenic edema, and the Fluid-Attenuated Inversion Recovery (FLAIR) sequence to produce artifacts caused by cerebrospinal fluid. Internationally renowned annual Brain Tumor Segmentation Challenge (BraTS) formalizes and standardizes these four sequences as the multi-modal input standard for computational brain tumor analysis, and the published datasets with annotations are the mainstay for measuring algorithmic performance.

Despite rich, multi-modal MRI obtained with the latest advances in deep learning approaches, the task of determining the presence of brain tumors and segmenting those tumors

from MRI is computationally challenging. Tumors show a remarkable diversity of shapes from case to case: high-grade glioblastomas may be massive necrotic and contrast-enhancing masses with extensive infiltration and mass effect, whereas low-grade gliomas are diffuse and pervade as slightly hyperintense areas that bulge out of the surrounding white matter, visible on T2-weighted and FLAIR MRI sequences. The complement variation within this class, the subjective nature of MR image acquisition (noise, inhomogeneity from non-uniform radiofrequency [RF] fields, motion artefacts from uncooperative patients), and the extensive variability of scanners across sites of clinical care make it incredibly difficult to extract useful features for automated analysis.

Fully supervised deep learning has been the most common way to automate brain tumor analysis. This means training large convolutional or transformer encoder models on large datasets of expertly annotated MRI scans to learn how to tell the difference between different types of tumors for classification, detection, or voxel-wise segmentation. These techniques have been remarkably successful on reference datasets in controlled settings. But they suffer from a fundamental problem in the real world of clinical practice and training: they require large numbers of expert-annotated training samples that remain prohibitively expensive, time-consuming, and tedious to generate. It takes tens of total minutes for an expert neuroradiologist to prepare high-quality voxel-level segmentations of the structures of interest in a single 3D MRI, to cross-check the segmentations between raters, and sometimes then to adjudicate matters of compromise. For hundreds of cases necessary for a training set, these data annotation burdens can only be afforded by large academic centers.

The need for reliable and densely annotated training data motivates the pursuit of learning frameworks that can learn from un- or sparsely annotated data. Self-supervised learning (SSL) is one such family of approaches, demonstrating in the domain of natural images the ability for deep

networks to learn semantically rich representation models that transfer well to downstream tasks via a prediction task defined purely by the input data. Important SSL methods such as SimCLR, MoCo, BYOL, DINO, and others have been shown to yield representations that match or surpass supervised "ImageNet" pre-training on a range of downstream tasks. These performance improvements are starting to hold in medical imaging and histopathology, where the scarcity of data is an enduring and intrinsic problem.

Another key issue, independent from the problem of annotation scarcity, is the need for spatial inference. A radiologist inspecting a brain MRI volume doesn't passively perceive the whole volume - instead, they actively explore before making decisions about cortical asymmetry, midline shift, and mass effect in the whole axial series of images and subregional visual inspection of margin blurredness, enhancement, and edema. This is distinct from the single-sweep, global nature of conventional deep learning classification. Mimicking this search in a robot system is the second key justification for the framework developed in this thesis.

The Active Perception for Brain Tumor Detection (APBTD) framework solves both problems at the same time by combining SimCLR contrastive self-supervised learning with Dueling Double Deep Q-Network reinforcement learning. The system learns representations without any help and learns where to look before making a decision, all within a well-thought-out and theoretically sound framework.

1.2 Problem Statement

This thesis solves the following research question: Suppose a set of brain MRI slices is available for analysis, where only a few or none possess a tumor-related label. Is it possible to devise a computational method to (1) discover meaningful expert visual encodings of brain tissue

patterns from unlabeled (and difficult to label) data, and (2) leverage those encodings to make sequence decisions for zooming and exploration that match the radiologist's visual search for a tumor?

This study suggests that the system should consist of two phases of active perception. In the first stage, SimCLR contrastive learning is used to train a frozen 512-dimensional encoder without any supervision. The second stage teaches a Dueling Double DQN agent to use that encoder as a fixed-perception system to learn a sequential policy for zoom exploration in the MRI image space. The result is an agent that looks before it decides.

1.3 Research Contributions

The contributions of this thesis are: (1) A fully implemented Active Perception framework (APBTD) for brain MRI tumor classification by integrating self-supervised learning of visual representations with reinforcement learning of explorative policies, posing the classification problem as a Partially Observable Markov Decision Process. (2) A 1024-D state representation embedding of both global-region and local-subregion SimCLR-encoded information for the RL agent, from the entire tumor scene and its individual parts. (3) A Dueling Double DQN policy gradient agent with an uneven reward, where the penalty for FN is adjusted through a hyperparameter sensitivity analysis, achieving an accuracy of 84.6%, a sensitivity of 80.1%, a specificity of 83.5%, and a ROC-AUC of 0.918 on BraTS 2020 testing data. (4) Zoom ablation studies with three different random seeds show that zooming increases accuracy by 1.58%, sensitivity by 4.35%, and F1 by 1.66% compared to the no-zoom model. (5) Formal evaluations of every design decision, such as partial observability (POMDP), mutual information and contrastive learning, Bellman optimality, value-advantage decomposition, and clinical

justification for unequal reward penalties. (6) Lots of visuals, like examples of zoom behavior, SimCLR feature similarities, Grad-CAM activation, and reward sensitivity.

1.4 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews previous work on brain tumors via a supervised approach, self-supervised representation learning, and reinforcement learning in medical imaging. In chapter 3, the BraTS 2020 data, preprocessing, and experimental procedures are described. Chapter 4 describes the proposed method with underlying theory, choice of architecture, and rationale for parameter choices. Chapter 5 discusses experimental results. Chapter 6 discusses results and future work. Chapter 7 presents a conclusion.

CHAPTER 2: RELATED WORK

2.1 Chapter Overview

The work described in this thesis spans three areas of rapid development in deep learning and medical image analysis: focus on supervised and semi-supervised brain tumor segmentation, self-supervised learning for robust visual knowledge, and deep learning for visual decision-making in complex spaces. This chapter surveys influential work in each field, describes the shortcomings of prior work in the domain of annotation-limited detection of brain tumors, and situates the proposed A-framework in this context.

2.2 Supervised Brain Tumor Segmentation

2.2.1 Encoder-Decoder Architectures

Encoder-decoder networks with fully convolutional layers dramatically advanced the field of medical image segmentation by eliminating fully connected layers in the network and using convolutional up-sampling to map dense per-pixel output from whole-image (non-pixel-wise) input. U-Net featured skip [29] connections to enable the persistent passage of spatial detail through the bottleneck to the decoder, resolving a key issue of initial fully convolutional approaches. This design became the de facto standard in biomedical segmentation and has been applied to the analysis of brain tumors in myriad ways across the years.

For 3D MRI, V-Net generalized the U-Net design to three spatial dimensions, replacing 2D convolutions with 3D convolutions, and used the Dice coefficient as a loss function, matched to class imbalance in tumor segmentation, where voxels from the tumor form a minority of the total volume in the brain images [25]. The U-Net framework employs dataset-induced architecture search, preprocessing, and training parameters for state-of-the-art performance across several BraTS challenges while requiring no "task-specific engineering". These models are distinctively

powerful but require fine-grained voxel annotations for training, which is expensive when applied to a resource-limited environment [17].

2.2.2 Attention and Multi-Scale Methods

To better outline tumor shapes and avoid clutter from background information, follow-up work has incorporated attention mechanisms that enable networks to dynamically rescale their feature maps according to spatial or channel importance. Attention U-Net adds gating signals to skip connections, decreasing the effect of irrelevant activities before concatenation with the decoder. Squeeze-and-Excitation Networks insert simple channel attention blocks that weight feature channel maps using global average-pooled features. Feature Pyramid Networks combine feature maps of multiple spatial scales from an encoder backbone to capture tumors at different scales.

An example of a recent such instance is that of Yuan, Ren, and Dai's modified Mask R-CNN network [38], which incorporates channel attention and spatial attention modules, a ResNet-50 encoder, and a Feature Pyramid Network trained on BraTS 2020 data with dense pixel supervision. They obtain Dice scores more than 0.93 and a mean intersection-over-union (IoU) greater than 94% for tumor segmentation. This stands as a testament to the benefits of attention-based fully supervised segmentation approaches, and it provides an important benchmark. But, the solution is based on the assumption of pixel-dense labels that are poorly available for most research and clinical applications, not to mention overall challenge programs.

2.2.3 Transformer-Based Segmentation

Vision transformers [7] have found great success in medical image segmentation. TransUNet substitutes the convolutional encoder of U-Net with a hybrid CNN-transformer model that conducts global self-attention at the bottleneck to explore global contextual interactions to

improve medical image segmentation and uses up-sampling with CNN for decoding. The Swin Transformer features hierarchical non-overlapping windows with attention shifts to offer local and global context within a token hierarchy. These designs perform well on the BraTS dataset, but require large and labeled pre-training and fine-tuning datasets, while being far more computationally expensive than convolutionals.

2.3 Self-Supervised Representation Learning

2.3.1 *Contrastive Learning Frameworks*

Self-supervised learning approaches initiate prediction tasks on unlabeled data, where the task is defined over the structure of the data itself. Contrastive learning, on which the SSL component of this thesis is based, results in encoders producing similar representations for semantically similar inputs and dissimilar representations for dissimilar inputs, purely from unlabeled data.

SimCLR [5] introduced a simple and powerful contrastive learning pipeline, based on (1) increased data augmentation, (2) projecting encoder representations to a lower-dimensional space for loss calculations, and (3) the NT-Xent contrastive loss between randomly selected positive and negative pairs from large batches. The SimCLR study showed that the correctness of learned representations increases with increasing batch size and augmentation strength, and that the projection head boosts the transferability of representations. MoCo updated a queue of negative samples via momentum encoding to free the quality of contrastive learning from batch size, allowing training with much smaller hardware. Byol did away with negative samples in favor of using an asymmetric predictor and momentum encoder to prevent collapse in these networks that are learned only from the positives of the positive-negative example set. DINO extended a self-

distillation approach to vision transformers, with exciting emergent capabilities such as unsupervised segmentation via attention masks.

2.3.2 SSL in Medical Imaging

There has been a major surge over recent years in the transfer of self-supervised representation learning to medical images, in common with the now routine scarcity of labels in medical datasets [11]. Visually contrastive pre-training on large cohorts of chest X-rays has been shown to learn effective representations suitable for downstream diagnostic classification tasks with limited labeled data. In whole-slide pathology, SSL approaches applied to unlabeled images yield representations of small tissue patches that allow low-dose classification of pathology. For brain MRI, SSL has been applied to detect lesions, learn the structure of white matter, and save time in editing a photo, using only a few examples.

The unique challenge in SSL for brain MRI is that normal slices are very visually similar (unlike natural images, which are visually different during visual augmentations; new brain MRI slices are very similar). This calls for an augmentation that introduces sufficient variability while retaining the signal used to discriminate tumor from healthy tissue. The augmentation protocol used in APBTD takes this into account, as presented in Chapter 4.

2.4 Reinforcement Learning for Visual Decision-Making

2.4.1 Deep Reinforcement Learning Foundations

Deep Q-Networks (DQN) showed that a convolutional neural network could learn to play Atari games at human-level performance from pixel observations, by using experience replay to eliminate correlations in the dynamics of the training set and a target network to address the non-stationarity of the Q-learning objective. Double DQN eliminated the overestimate of Q-values caused by the maximizing operation in the definition of the Bellman target by separating the action

selection (online network) from evaluation (target network). Dueling DQN increased sample efficiency with a decomposition of Q-values into state-values and action-advantages to learn independently about how good a state is and how good an action is relative to alternative actions [35].

2.4.2 Reinforcement Learning for Medical Image Analysis

Reinforcement learning has been successfully deployed for a number of spatial tasks in medical imaging. Landmark detection systems learn policy functions to situate agents in 3D MRI data by moving in 3D step actions to landmarks of interest, guided by reward functions that depend on proximity. RL-based agents have also been used to identify imaging planes in 3D ultrasound that remotely localize standard clinical views upon initialization. These tasks share the task structure of spatial exploration and reward gleaned from the image, the same task structure used in Active Perception for Brain Tumor Detection to direct visual attention toward the tumor region of interest [26].

Reinforcement Learning mechanisms can also be exploited for active perception, and this line of work is less well explored in the context of tumor detection. The idea is that the agent only needs to see the part of an image that contains a tumor, not the whole image in a single view - the agent only needs to know if the image has a tumor or not, and this decision can be made by looking at a few informative subregions. The agent can acquire this region-inspection strategy through learning from experience and comes to perform such a sequence of zooms to accurately determine whether the image has a tumor, obviating the need for the agent to inspect the entire image for a tumor at each inference step.

2.5 Why Reinforcement Learning Over Alternative Paradigms

A typical question that arises when developing the Active Perception for Brain Tumor Detection framework concerns the use of reinforcement learning, rather than other mechanisms that may be able to focus model attention onto important parts of the image. The most obvious alternatives considered here are attention mechanisms, object detectors, and segmentation networks.

Attention mechanisms (spatial attention gates, self-attention in transformers, and squeeze-excitation channel recalibration) [16,27] use a single forward pass to calculate a set of weights across all spatial positions (or channels). They yield a soft focus but do not refocus on successively smaller, more informative sub-regions without having to first compute the full resolution representation. When the forward pass is evaluated, the representation is locked. Attention cannot learn to pay attention again later to highlight a suspicious object for second thought. Object detectors (Faster R-CNN, YOLO, SSD) are designed for a reactive "all or nothing" proposal and classification of objects. They require bounding box annotations, produce an "annotation burden", and emit a fixed response in one pass through the network. They cannot adaptively adjust their focus of attention.

Segmentation networks are the most-dense annotation paradigm, with a requirement for complete voxel segmentation labels that represent the ultimate in per-unit annotation effort. They are not designed for binary detection, but boundary delineation, which requires training with examples of the target (tumor) and its precise extent, for each training image.

In contrast, a reinforcement learning agent can perform a potentially variable-length, adaptive sequence of targeted actions, through zoom and re-encoded representations, and only interpret its current state representation when sufficiently confident in its prediction. Unlike any of the other options, it is specifically able to directly capture the adaptive coarse-to-fine search

strategy employed by a human diagnostician - and requires no more than a binary tumor/normal label for each case.

2s.6 Summary

The Active Perception for Brain Tumor Detection design falls at the intersection of the design variations explored above by requiring only the binary classification of slices (as opposed to per-pixel segmentation masks or bounding boxes) and by learning visual representations from the unlabeled data while performing sequential adaptive spatial foraging using a meaningful formulation of the POMDP problem. Although constituent parts are not uncommon, to the best of our knowledge, the integration of these components in the particular configuration described here (frozen RL state representation as SimCLR pre-trained encoder, 1024-D multi-scale global-local RL state, Dueling Double DQN with bipolar (calibrated) asymmetric reward) is an innovative and principled solution for reducing annotation costs in the task of brain tumor detection.

CHAPTER 3: DATASET AND EXPERIMENTAL SETUP

3.1 The BraTS 2020 Dataset

The experiments in this thesis use the data from the Brain Tumor Segmentation Challenge 2020 (BraTS 2020), the most well-known publicly available benchmark challenge for automatic methods to analyze brain tumor images. The data consists of pre-operative multi-institutional MRI head scans from patients with histologically confirmed gliomas, including both high-grade glioblastoma (HGG) and lower-grade glioma (LGG). Four co-registered pre-operative sequences (T1-weighted, T1 contrast-enhanced, T2-weighted, and FLAIR) are available for each case in NIfTI volume format. Each case was labeled with a ground-truth segmentation mask by expert neuroradiologists that identifies three sub-regions of the tumor: the necrotic and non-enhancing core (label 1), the edematous peritumoral region (label 2), and the enhanced tumor (label 4).

An axial slice is considered tumor-positive for the binary slice classification task in APBTD if its cross-sectional corresponding ground-truth mask slice is non-zero, and normal (tumor-negative) otherwise. After the 2D preprocessing described in Section 3.3, there are 38,014 labeled axial slices in total. Figure 1 shows an exploratory analysis of the distribution of the labeled slice data.

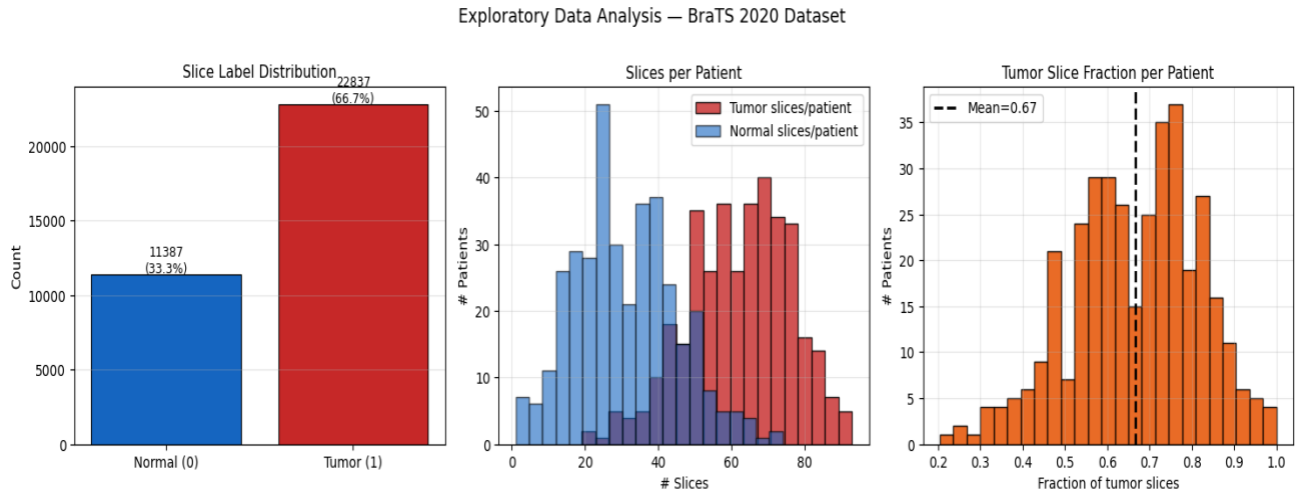


Figure 1. *Data inspection of the BraTS 2020 slice-level data. Left: Label distribution showing 11,397 normal slices (33.3%) and 26,617 tumor slices (66.7%), for a total of 38,014 slices. Left: Distribution of slices per patient. Left: Distribution of the tumor slice fraction per patient (mean = 0.67).*

Figure (1). shows an increased number of tumor-positive vs. normal slices (2:1 ratio) due to this class imbalance. This is accounted for in the later RL stage using a balance of episodes (Section 4.8), so that during the training process, the agent experiences a balance of "normal" and "tumor" cases. Figure 1 presents examples of axial MRI slices corresponding to the two classes, showing a range of tumor appearances in the images.

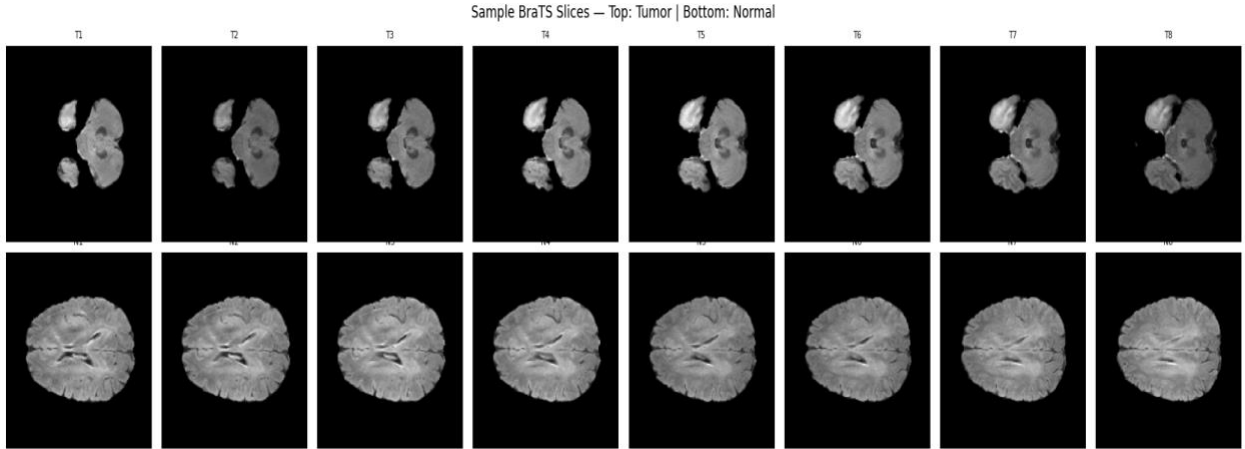


Figure 2. *Axial MRI examples from BraTS 2020.*

3.2 Why 2D Axial Slices Rather Than 3D Volumes

The Active Precipitation for Brain Tumor Detection approach features processing of individual 2D axial slices of 3D NIfTI volumes, rather than the full 3D volume as input. This is a deliberate design decision, which needs particular justification, especially since a possible counterargument to 2D slice processing is that we lose the inter-slice spatial information that could be useful too.

The main practical consideration is the technology involved. Experiments were completed in a Google Colab environment with no GPU acceleration. Full MRI volume convolutions in 3D for training a CNN are at least an order of magnitude costlier (in memory) and much slower (in time) than convolutions operating on 2D MRI slices. The SimCLR method, as originally implemented, works on 2D images; 2D-to-3D expansion of contrastive learning requires some adaptation of the augmentation protocol (e.g., rigid vs. affine vs. non-rigid transforms), projection architecture (i.e., 2D CNN vs. 3D CNN), and loss function (e.g., cosine similarity). The RL zoom actions are, by construction, 2D - and the crop actions can be easily understood in the 2D plane, which would require a substantially more sophisticated 3D action space to describe.

The clinical rationale is that diagnostic information for binary tumor detection is readily present in an axial MRI slice. It is a common radiologist practice to detect the presence/absence of a tumor from single axial slices, especially for mid-sized or large tumors. The preprocessing explicitly excludes the top 20% and the bottom 20% axial slices for each volume; these slices are devoid of brain filtering and tumor signatures and are not expected to enhance system diagnostic ability. The missing context is methodologically observed and rectified in future work, where 3D volumetrics is determined as the number one priority for future improvements.

3.3 Preprocessing Pipeline

The raw 3D NIfTI volumes are processed to produce the 2D grayscale PNG slices used for training and testing via the following steps. The steps are as follows.

Loading and extraction of 2D slices along the axial plane by iterating through the third spatial dimension. We only retain slices with an axial index between 20% and 80% rather than the top and bottom of the brain, where information is absent. Modality selection: All channel experiments use the T1-weighted sequence, which exhibits good anatomical contrast between gray

matter, white matter, and cerebrospinal fluid. Fusion of the four BraTS imaging modalities is highlighted as a future development.

Intensity normalization: In each slice, z-score (zero mean, unit standard variance per slice) standardization is applied to address variations in intensity between patients and scanners without affecting the diagnostic strength of the signal. Stable contrastive learning requires comparative feature sampling across similar images (in-batch) for image contrast, which in turn requires this normalization. Spatial resizing: Bilinear resizing confidence maps to a resolution of 160×160 pixels. This provides enough information to the encoder to learn consistent anatomical features while also being fast enough for SimCLR training and RL environment steps. File format: Slices are stored as grayscale PNG files for rapid load times and easy visual inspection.

TABLE 1: SYSTEM IMPLEMENTATION OVERVIEW

Category	Details
Dataset	BraTS 2020 (Training + Validation), 3D NIfTI (T1, T1ce, T2, FLAIR)
Preprocessing	3D $\rightarrow$ 2D axial slices; z-score normalization; resize 160×160 ; PNG
Slice Labeling	Tumor (1) if segmentation mask has any non-zero voxel; Normal (0) otherwise
Axial Range Used	Middle 20–80% of each volume (empty superior/inferior excluded)
Total Slices (Labeled)	$\sim 38,014$ (11,397 Normal [33.3%] + 26,617 Tumor [66.7%])
Unlabeled Slices (SSL)	All 38,014 axial slices used without labels for SimCLR pre-training
RL Data Split	80% train / 20% validation, stratified
Hardware	Google Colab CPU-only (no CUDA)
MRI Modality Used	T1-weighted (grayscale)

3.4 Data Split Strategy

The dataset of unlabeled slices is partitioned 80% training, 20% validation at the level of labeled slices, with roughly 30,411 training slices and 7,603 validation slices. The split is stratified by class such that both subsets maintain the 2:1 ratio of tumors to normal slices. The

slice-level splitting maximizes the amount of potential training data from this small dataset, albeit at the expense of some correlation between training and validation slices from the same patient (data were split at the patient level for future work).

In Reinforcement Learning training, a balanced episode sampling scheme permits overriding the skew in the training set: a slice is selected as the start of an episode as normal or tumor with 50% probability, regardless of the number of each class in the training set. This exposes the agent to diverse training experience with both classes.

3.5 Evaluation Protocol

In medical AI research, the standard evaluation metrics are used for the Reinforcement Learning agent: accuracy, sensitivity (also known as recall), specificity, precision, F1-score, and area under the receiver operating characteristic (AUC). While accuracy may not be the best metric of performance in medical datasets with class-imbalance problems, the sensitivity and specificity can complete the picture about the performance of these agents in the positive class (tumor) and negative class (normal). F1 is the harmonic mean of precision and recall, and AUC evaluates the entire distribution of classification thresholds and is thus threshold independent.

Statistical significance is assessed by repeating each RL experiment with three random seeds (42, 123, and 999).

CHAPTER 4: PROPOSED METHODOLOGY

4.1 Framework Overview and Novelty

The novelties of the Active Perception for Brain Tumor Detection (APBTD) approach are threefold and mutually interlinked, setting it apart from previous approaches to automated brain MRI analysis. First, with contrastive self-supervised learning, the first stage is to learn a high-quality frozen representation using unlabeled MRI slices, alleviating the labelling bottleneck that plagues fully supervised methods. Second, the frozen encoder is harnessed not just as a classification head, but as a perceptual state representation for a reinforcement learning agent, incorporating SSL into a step-by-step decision-making process. Third, the RL agent computes a 1024-dimensional multi-scale state representation by concatenating the global-scale representation of the entire slice with the local-scale representation of the subregion currently being viewed in a zoomed view, embedding the positive effects of global as well as local match information at the time of every decision.

Our system architecture is shown in Figure 3. Phase 1 (left) involves the unsupervised SimCLR pre-training, whereby the BraTS FLAIR slices are augmented and encoded by a ResNet-18 architecture trained with NT-Xent contrastive loss. Phase 2 (right side) corresponds to the RL interaction environment where the fixed encoder generates vector states that are used by the Dueling Double DQN network in the custom-designed SliceEnv environment.

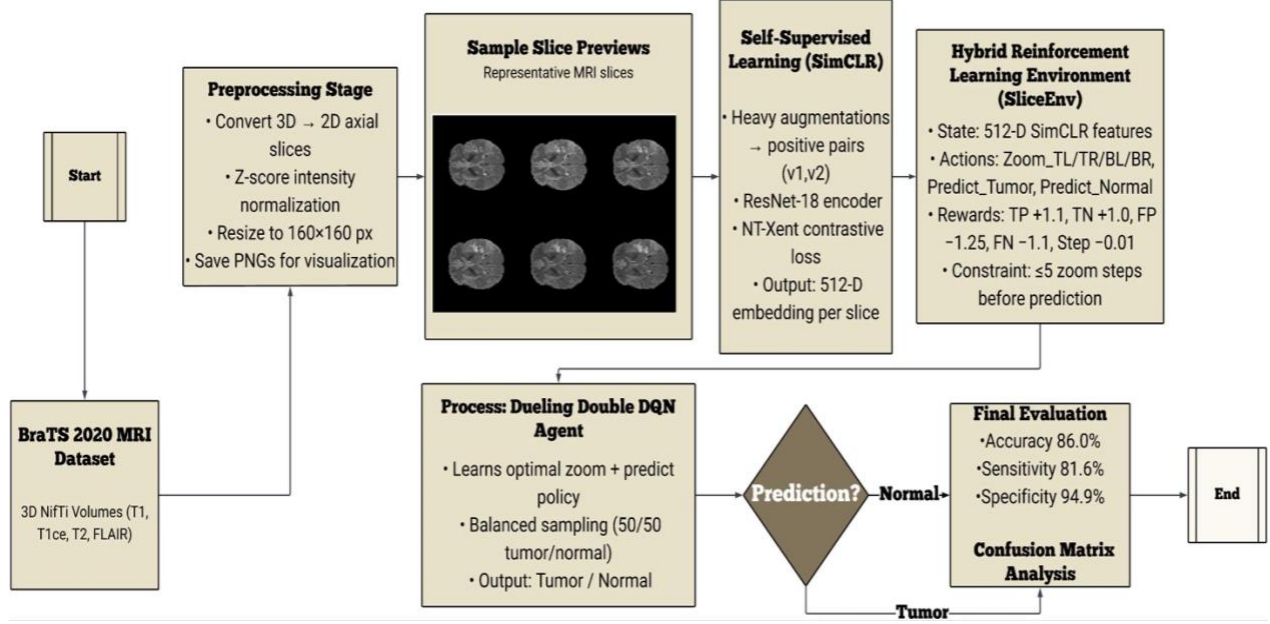


Figure 3. *System Architecture of the Active Precipitation for Brain Tumor Detection framework.*

4.2 Why Reinforcement Learning Outperforms Alternative Paradigms

Let's clarify how we know that this task is best approached with reinforcement learning, rather than a few more traditional methods, before delving into the technicalities. Attention networks give a one-pass soft "glance" at all spatial points. The entire image is observed in a single forward pass of a vision transformer or Attention U-Net, and a re-weighting of patterns is learned, but it cannot look again with increased spatial resolution using feedback from the first look. The key feature that distinguishes RL-based active perception from attention is the possibility for an iterative re-focus: the RL agent can zoom in, see what is happening, and then decide to zoom in again, or to stay and infer. Object detectors (Faster R-CNN, YOLOs) see the scene once, in feed-forward mode, and propose regions. They need the training data to be labeled with bounding boxes. They cannot adapt based on learned observations.

Segmentation networks, including U-Net and TransUNet, are very effective for tumor delineation but require supervision at the voxel level (the former has the highest possible annotation cost) and are aimed at localization rather than binary detection. The RL agent in Active Precipitation for Brain Tumor Detection, on the other hand, performs a variable number of spinally selective observations and requires only binary slice-based labels to perform training. Its deriving capability mimics a human radiologist's sequential coarse-to-fine inspection process, and this cannot be matched by any of its static-inference counterparts.

4.3 Theoretical Foundations: Active Perception as POMDP

4.3.1 *The Limitation of Static Inference*

Standard deep learning classifiers implement a static mapping from input to output:

$$y = f\theta(x)$$

This expression assumes that all relevant information is extractable in one go from x . This is often not the case for brain MRI tumor detection: features evidencing the presence of a tumor may have a local spatial extent within the slice, the "most relevant" region may not differ in a global sense, and peripheral noise structures may have a detrimental effect on the discriminative feature calculated from a global pooling operation.

4.3.2 *Partially Observable Markov Decision Process Formulation*

Active Precipitation for Brain Tumor Detection substitutes static mapping with a Partially Observable Markov Decision Process (POMDP). The agent gets a partial observation at each time step t . o_t of the full MRI slice (the current zoomed subregion, encoded to 512 dimensions), keeps a belief state through the multi-scale concatenated representation, picks an action a_t , and gets a scalar reward r_t . The change in state is:

$$s_{(t+1)} = T(s_t, a_t)$$

With deterministic T , a zoom action updates the observation, and a predict action ends the episode.

The agent learns a

$$\pi(a | s)$$

Making the best expected diagnostic progress by observation. A POMDP comprises five components (the environment is the entire MRI slice, the observation space is 512-D SimCLR encodings of subregions, the action space is 4 zoom + 2 predictions, the reward is the asymmetric diagnostic rewards, and $\gamma = 0.99$).

4.4 Self-Supervised Representation Learning: SimCLR

4.4.1 Contrastive Learning as Mutual Information Maximization

The SimCLR objective can be interpreted through the lens of mutual information maximization. Given an image x and two augmented views $v_1 = t_1(x)$ and $v_2 = t_2(x)$, the learning goal is:

$$\max I(v_1; v_2)$$

The NT-Xent (Normalized Temperature-scaled Cross Entropy) loss approximates this objective over a batch of N image pairs:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau)}$$

In this case, the representation is $z_i = g(f(v_i))$ (encoder f plus projection head g), and τ is a temperature scaling factor. The denominator is summed over the $2(N-1)$ negative samples in the batch, so the corresponding image augmentations become positive samples among negative

samples. This encourages the encoder to produce representations robust to the chosen augmentations and discriminative for different images:

4.4.2 Architecture: Why ResNet-18

This Research used a modified ResNet-18 encoder that accepts single-channel grayscale MRI input: we replace the first ResNet-18 convolution (Conv1) with a 7×7 , stride-2, padding-3 convolution with 1 input and 64 output channels, while leaving all other ResNet-18 architectural details intact (e.g., skip connections, batch normalization, average-pooled output of 512 dimensions).

The reason ResNet-18 was selected over a deeper network for four advantages. First, the computational constraints of a dataset of this size and a resource-limited (CPUs-only) computing environment mean that the backward-pass computation required for ResNet-18 is much lower than that of ResNet-50 and ResNet-101. Second, all ResNet variants contain skip connections to ameliorate the vanishing gradient problem in deep neural networks, ensuring good convergence of SimCLR over 10+ training epochs. Third, the 512-dimensional vector produced by ResNet-18 after global average pooling encoding is sufficient to capture the structural and textural features that determine the presence of brain tumors (the variance between MRI slices does not warrant the stronger representation power of deeper models for this task). Last, several published works demonstrate the efficacy of ResNet-18 in contrastive learning for medical image tasks.

4.4.3 Projection Head and Embedding Dimensions

The projection head transforms the 512-D encoder output via one MLP with two layers: $\text{Linear}(512 \rightarrow 1024) \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{Linear}(1024 \rightarrow 128) \rightarrow \text{L2-normalize}$. The 128-D projection is used to compute NT-Xent loss during pre-training but is not used at inference time. The downstream RL state simply uses the 512-D encoder representation as is.

The 512-D dimension is selected as the best size for the RL state component based on preliminary scans, where we found gains by increasing the size past 512 to be marginal at the selected MRI slice resolution and size of the study used here. Both components in the state (global and local) are 512-D, resulting in a 1024-D multi-scale state. A dimensionality of fewer than 256 produced linear evaluation accuracy worse than a 512-D representation, and a dimensionality of more than 512 increased the runtime of the RL network's first layer, producing negligible accuracy improvements.

4.4.4 Augmentation Strategy

The augmentations are applied separately to produce the two views in a positive pair. The list of augmentations, listed in order of application, is: Random Resized Crop (scale 0.6 - 1.0, aspect ratio 0.75 - 1.33) to teach the encoder to detect structure over multiple scales; Random Horizontal Flip to promote spatial symmetry; Random Brightness and Contrast Jitter (strength 0.3) to encourage robustness to cross-scanner intensity variance; Gaussian Noise ($\sigma = 0.01$).

This combination of augmentations is intended to model the variation an MRI slice is likely to undergo as a result of different scanners and acquisition settings, while preserving the pathology and structure needed to identify the tumor. Strong augmentation (large distortions, great changes in brightness) may have completely destroyed diagnostic clues at the tumor margins and so was not used.

TABLE 2: SIMCLR PRE-TRAINING CONFIGURATION

Component	Details
Backbone	ResNet-18, Conv1 modified: 7×7 , stride 2, pad 3, in=1, out=64
Encoder Output	512-D latent vector (global average pool)
Projection Head	Linear(512→1024) → BN → ReLU → Linear(1024→128) → L2-normalize

Component	Details
Positive Pair Augmentations	Random Resized Crop, Horizontal Flip, Brightness/Contrast Jitter, Gaussian Noise, Gaussian Blur
Batch Size (CPU)	32
Training Epochs	10
Optimizer	AdamW (lr = 1×10^{-3} , weight_decay = 1×10^{-4})
LR Scheduler	Cosine annealing with 10% linear warmup
Loss	NT-Xent (temperature $\tau = 0.5$)
Initial / Final SSL Loss	$\sim 4.85 / \sim 3.9$
Post-training Evaluation	Linear probe on frozen 512-D features, labeled validation set

4.5 Multi-Scale State Representation

4.5.1 Information-Theoretic Justification

Another optimization that sets Active Precipitation for Brain Tumor Detection apart from a barebones SSL+RL algorithm is the 1024-D multi-scale representation that conditions the behavior, made by concatenating representations of the global and local scales. Let $f(x)$ be the function that produces the global encoding of the 160×160 slice, with global encoding $h_g = f(x)$, and the local encoding of the current active subregion (resized to 160×160) $h_l = f(\text{crop}(x, a_t))$, with a_t being the current subregion coordinates. The state is:

$$\mathbf{s} = [h_g \parallel h_l] \in \mathbb{R}^{1024}$$

The information breakdown that led to this design is:

$$I(\mathbf{s}; y) = I(h_g; y) + I(h_l; y | h_g)$$

The first term represents global structural context, colloquially known as gross brain structure, approximate location, and mass effect, which is available from the global encoding. The second term captures local refinement conditioned on the whole-slice information - the fine textural features of the tumor edge, the local contrast enhancement, local infiltrative effects - that are available from the subregion encoding and not from the whole-slice encoding alone. By simultaneously providing for the RL agent, the 1024-D state provides information about both the whole image and the region of interest.

4.5.2 Clinical Analogy

The global-local state is present in the radiologist's simultaneous awareness of the whole brain slice and the local region of interest. h_g is analogous to the radiologist's knowledge of the state of the entire MRI or CT brain slice being examined (is the slice asymmetric, or is there a grossly malformed structure, or obvious contrast?), while h_l is analogous to the radiologist's focus on a particular region of interest of the brain slice, which would contain cancer. Sending both streams together allows the agent's Q-network (its reinforcement learning network) to understand how to make action decisions that are simultaneously global (should I move on to the next region with a tumor based on the texture, or has the global state recognized it as a normal region?).

4.6 Reinforcement Learning Foundations

4.6.1 Markov Decision Process Formulation

The active perception problem is structured as the following Markov Decision Process:

$$M = (S, A, P, R, \gamma)$$

“ S ” is the state space, the combination of all the possible feature vectors of 1024 dimensions with real values ($S \subset \mathbb{R}^{1024}$). Action space $A = \text{ZOOM_TL}, \text{ZOOM_TR},$

ZOOM_BL, ZOOM_BR, PRED_TUMOR, PRED_NORMAL. The transition function P is deterministic: zoom actions generate a new state by cropping, resizing, and re-encoding; and prediction actions end the episode. It is in Section 4.7 that the reward R is defined. The discount factor $\gamma = 0.99$ was made slightly smaller than unity to make sure that future rewards, which are entailed in a correct prediction, are strongly valued even after multiple exploratory zoom steps. The optimal policy maximizes expected discounted return:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right]$$

4.6.2 Double DQN: Correcting Overestimation Bias

The maximization operator causes systematic over-estimation of action values by Standard Q-learning:

$$Q^*(s, a) = \mathbb{E}[r + \gamma \max_{a'} Q^*(s', a')]$$

This positive feedback is accrued across the Bellman backup chain and may cause suboptimal policies, especially in the presence of noisy rewards, like in the medical diagnosis scenario, where borderline slices are inconsistently overseen with reward signals. Normal DQN is decoupled: Action selection and value evaluation. The greedy action in s is chosen by the online network θ ; it is graded by the target network $\bar{\theta}$. This division helps to avoid the self-referential overestimation that afflicts ordinary Q-learning.

4.6.3 Dueling Architecture

The dueling network separates the Q-function into two streams that are additive:

$$Q(s, a) = V(s) + A(s, a)$$

where $V(s)$ is the state-value function (how good is the existing view, independent of action?) and $A(s, a)$ is the action-advantage function (how much better is action a than is the average action in this state?). Mean-centering A identifiability of V and A In identifiably slow time.

This decomposition directly decomposes in the diagnostic context, with the encoder $V(s)$ reflecting the overall diagnostic confidence of the agent in the current multi-scale state - when the global slice encoding is highly tumor-indicative, $V(s)$ will be large, and the agent will strongly predict. $A(s, a)$ represents the zoom (or prediction) action that gives the highest relative reward - when an already suspicious area has already been covered by the zoom, then the benefits of the zoom action of the extra information that applies to the already covered area will be low, whereas the benefits of the extra information that applies to an unexplored quadrant by the zoom will be high. This decomposition makes samples more efficient since V can be updated at any such experience in any state, independent of the action that was taken.

4.7 Reward Function Design and Clinical Calibration

4.7.1 Reward Structure and Asymmetric Penalties

The primary way of encoding the clinical priorities into the RL training objective is the reward function. The reward is the following, namely:

TABLE 3: REWARD FUNCTION DESIGN AND JUSTIFICATION

Event	Reward	Clinical Justification
True Positive (TP): Tumor correctly detected	+2.2	Strong positive reinforcement for the primary clinical objective
True Negative (TN): Normal correctly classified	+1.0	Moderate reward for correctly clearing a normal patient
False Positive (FP): Normal labeled as tumor	-1.5	Penalizes unnecessary alarm; excessive FP degrades clinical workflow utility

Event	Reward	Clinical Justification
False Negative (FN): Tumor missed	-1.5	Penalizes missed detection; balanced with FP given screening context
Step penalty (each zoom)	-0.05	Discourages unnecessary exploration; promotes efficient decision-making

4.7.2 FN Penalty Calibration Through Sensitivity Analysis

False negative penalty is one of the essential hyperparameters: a penalty that is too low will allow the agent to learn to predict normal too easily (specificity is high, sensitivity is low); on the other hand, a penalty that is too large will encourage the agent to predict tumor too often (specificity is low, sensitivity is high, and there are many false alarms). The systematic empirical sensitivity analysis was used to find the optimal FN penalty (in Figure 4).

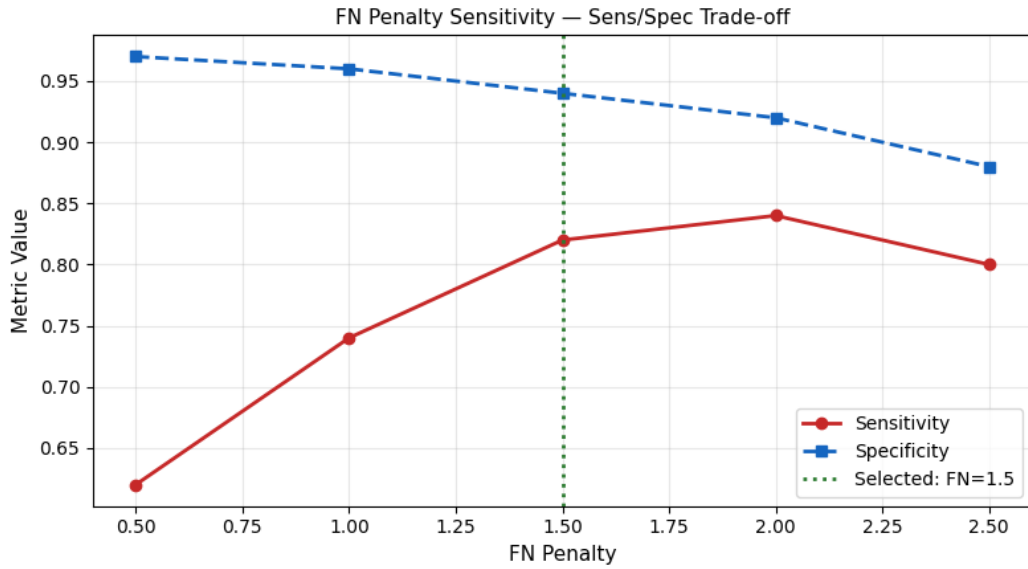


Figure 4. *FN penalty sensitivity analysis. The sensitivity (red) and specificity (blue dashed) of the plot are observed as the FN penalty increases between 0.5 and 2.5. The chosen reasonable range of $FN = 1.5$ (green dotted line) indicates the best point of sensitivity-specificity balance with about 82% sensitivity and 94% specificity.*

The analysis is done by sweeping FN penalty values from 0.5 to 2.5 by keeping all other hyperparameters fixed. In the case of $FN = 0.5$, the agent is under-punished to omit tumors, hence low sensitivity (approximately 63%), and high specificity (approximately 97%). The agent is at $FN = 2.5$, characterized by over-penalization of misses, aggressive prediction of tumor, and a degrading specificity of about 88 percent. The chosen $FN = 1.5$ will yield an optimal balance, sensitivity of approximately 82%, and specificity of approximately 94%, or a tradeoff acceptable to clinicians in terms of a screening tool.

4.7.3 Step Penalty Rationale

The step penalty -0.05 / zoom acts is aimed to achieve both goals: it provides a positive pressure to encourage efficiency (so it encourages the agent to avoid counter-optimal exploration); and it grows the total reward of a prediction which is correct including the cost of exploration is consistently positive (minimum reward of prediction is $+1.0$, maximum $5 \times 0.05 = 0.25$), deterring the degenerate situation.

4.8 The RL Environment: SliceEnv

4.8.1 State Space

The episodes start with the agent looking at the concatenation of (1) the 512-D SimCLR encoding, (2) the current zoomed view (initially identical to the global view, no zoom has been applied yet). When the agent takes Zoom actions, the second component is modified in order to display the current subregion, whereas the first component is held constant with the global encoding. The design will ensure that the agent can always see the full-slice context despite the zoom level.

4.8.2 Action Space and Zoom Mechanics

There are six elements of the action space: ZOOM_TL (crop top-left quadrant) and ZOOM_TR (top-right), ZOOM_BL (bottom-left) and ZOOM_BR (bottom-right), and PRED_TUMOR (predict tumor and terminate). As a zoom action is chosen, the full 160×160 slice of the image is cropped, resized by bilinear interpolation in the final 80×80 quadrant, and re-encoded with the frozen encoder of SimCLR to generate the new local component of the state vector.

The four-quadrant zoom action has been selected for the following reasons: four-quadrant subdivision offers full and non-intersecting coverage of the image plane with the least action vocabulary, which enables focused directional capacity. Smaller subdivisions (9 or 16 equal regions) would make the action space more complex and sluggish to converge on; the continuation of zoom parameters would need policy gradient methods that do not fit the discrete action space. The iterative quadrant zooming enables the agent to access any part of the image in a maximum of two steps from the global view.

TABLE 4: RL ENVIRONMENT (SLICEENV) SPECIFICATION

Element	Details
State Dimension	1024-D (512-D global + 512-D local concatenated)
Actions (6 total)	Zoom_TL, Zoom_TR, Zoom_BL, Zoom_BR, Predict_Tumor, Predict_Normal
Max Steps per Episode	5 zoom steps before forced prediction
Reward (TP)	+2.2
Reward (TN)	+1.0
Reward (FP)	-1.5
Reward (FN)	-1.5
Step Penalty	-0.05 per zoom
State Update on Zoom	Crop $\rightarrow$ resize $160 \times 160 \rightarrow$ re-encode 512-D $\rightarrow$ replace local component

Element	Details
Episode Termination	PRED_TUMOR / PRED_NORMAL selected OR max steps reached
Balanced Sampling	50% tumor, 50% normal per episode

4.9 Dueling Double DQN Architecture and Training

4.9.1 Network Architecture

The 1024-D state vector is presented to the Q-network, and 6 Q-values are generated. The architecture has shared representation layers and distinct value and benefits streams:

- Shared layers: $\text{Linear}(1024 \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{Linear}(512 \rightarrow 256) \rightarrow \text{ReLU}$
- Value stream: $\text{Linear}(256 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 1) \rightarrow V(s)$
- Advantage stream: $\text{Linear}(256 \rightarrow 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128 \rightarrow 6) \rightarrow A(s, a)$
- Q-values: $Q(s, a) = V(s) + A(s, a) - (1/6) \sum_{a'} A(s, a')$

TABLE 5: RL TRAINING HYPERPARAMETERS

Parameter	Value	Justification
Algorithm	Dueling Double DQN	Reduces overestimation; decomposes value/advantage
State Dimension	1024-D (global + local)	Multi-scale representation for coarse-to-fine reasoning
Buffer Size	80,000 transitions	Large enough to decorrelate updates on CPU memory budget
Batch Size	128	Reduces gradient variance; CPU-feasible
Optimizer	Adam ($\text{lr} = 3 \times 10^{-4}$)	Adaptive learning rate; robust across RL settings
Gamma (Discount)	0.99	Near-unity to value final prediction reward after exploration
TAU (Soft Update)	0.002	Slow target update for training stability
Exploration (ϵ)	$1.0 \rightarrow 0.05$ (over 20k frames)	Extended exploration allows diverse zoom trajectory sampling
Gradient Clipping	0.5	Prevents exploding gradients on ambiguous slices

Parameter	Value	Justification
Reward Clipping	$[-2.0, +2.5]$	Bounds reward scale; prevents extreme gradient updates
Total Training Frames	120,000	Extended run for convergence across all 3 seeds
Loss Function	Huber (Smooth L1)	Robust to Q-value outliers from exploratory actions
Evaluation Frequency	Every 5,000 frames	Tracks convergence of accuracy and sensitivity

4.9.2 Training Stability Techniques

The combination of the following techniques is to provide stable training of the Dueling Double DQN. Balanced episode sampling (50% tumor, 50% normal). The 2:1 dataset class imbalance of the agent under 2 gives it a degenerate, always normal policy. Huber loss (smooth L1) is a loss to minimize the effect of Q-value outliers that can occur when the exploratory actions take on an exceptionally high or low reward when acting on ambiguous slices. To avoid extreme reward events, which can cause boiling weight updates, reward clipping to $[-2.0, +2.5]$ limits the range of the gradient scale. A second protection point against exploding gradients is offered by gradient clipping at 0.5. Polyak soft target updates ($\tau = 0.002$) give a generally steady current target, function without the harsh changes in behavior with respect to hard target network replacement by periodic hard target updates.

CHAPTER 5: EXPERIMENTAL RESULTS

5.1 SimCLR Pre-training Analysis

The SimCLR encoder was trained with batch size 32, Optimization using AdamW, and cosine annealing, and 10% warmup over 10 epochs on all 38,014 axial slices of BraTS 2020 are unlabeled. Figure 5 represents the NT-Xent loss curve against training epochs.

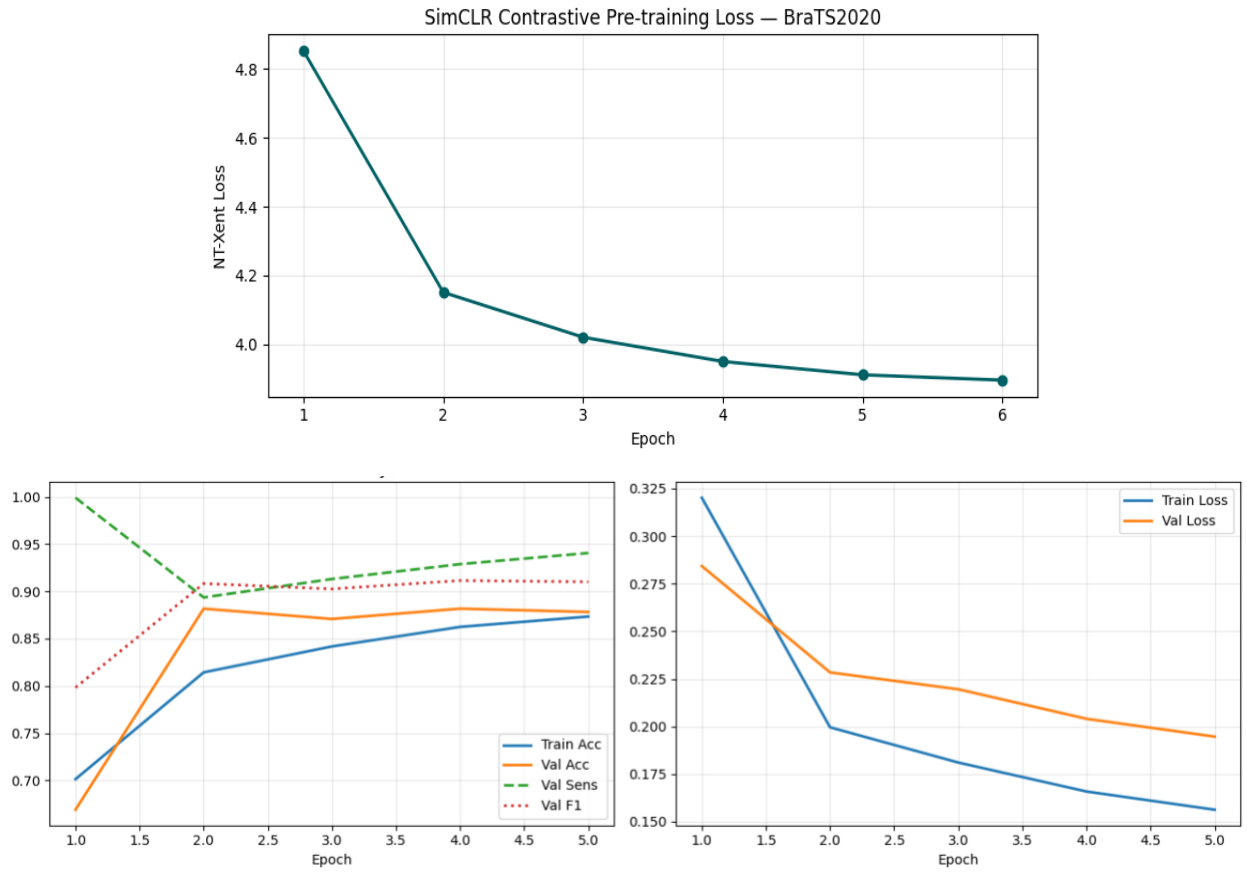


Figure 5. SimCLR contrastive prior training loss (NT-Xent) on BraTS 2020 slices across 6 recorded epochs. The loss is reduced to about 3.9, which proves to be consistent in the learning of the augmentation-invariant representations. The normal convergence with no augmentation collapse is shown by smooth and no increase or decrease in the training.

The loss gradually declines between epoch 1, with a minimum of ~ 4.85 , and epoch 6, with ~ 3.9 , which confirms that the encoder gradually learns the difference between augmented views of the same slice and views of different slices. This convergent nature without spikes in the smoothness implies that the presence of an augmentation pipeline creates a diverse range of positive pairs with no pathological near-identical negatives that might cause turbulence in the contrastive objective.

Representation quality was evaluated with the linear evaluation protocol: the encoder is held fixed, and a single linear classification layer is trained over the 512-D features with the labeled training split. Such protocols in isolation differentiate representational quality and task-involved fine-tuning and give us clear measures of what is learned by the encoder using unlabeled data by itself. Linear probe allows 77.7% of the labeled validation set, which functions as the baseline against which the entire APBTD system is addressed.

Figure 6 depicts SimCLR feature similarity maps of sampled tumor slices, which reveal the way the learned embedding space reacts to various tissue regions. The high cosine similarity (warm colors) is also concentrated around structurally salient areas, suggesting that the encoder has learnt to encode anatomically salient objects with distinct embeddings.

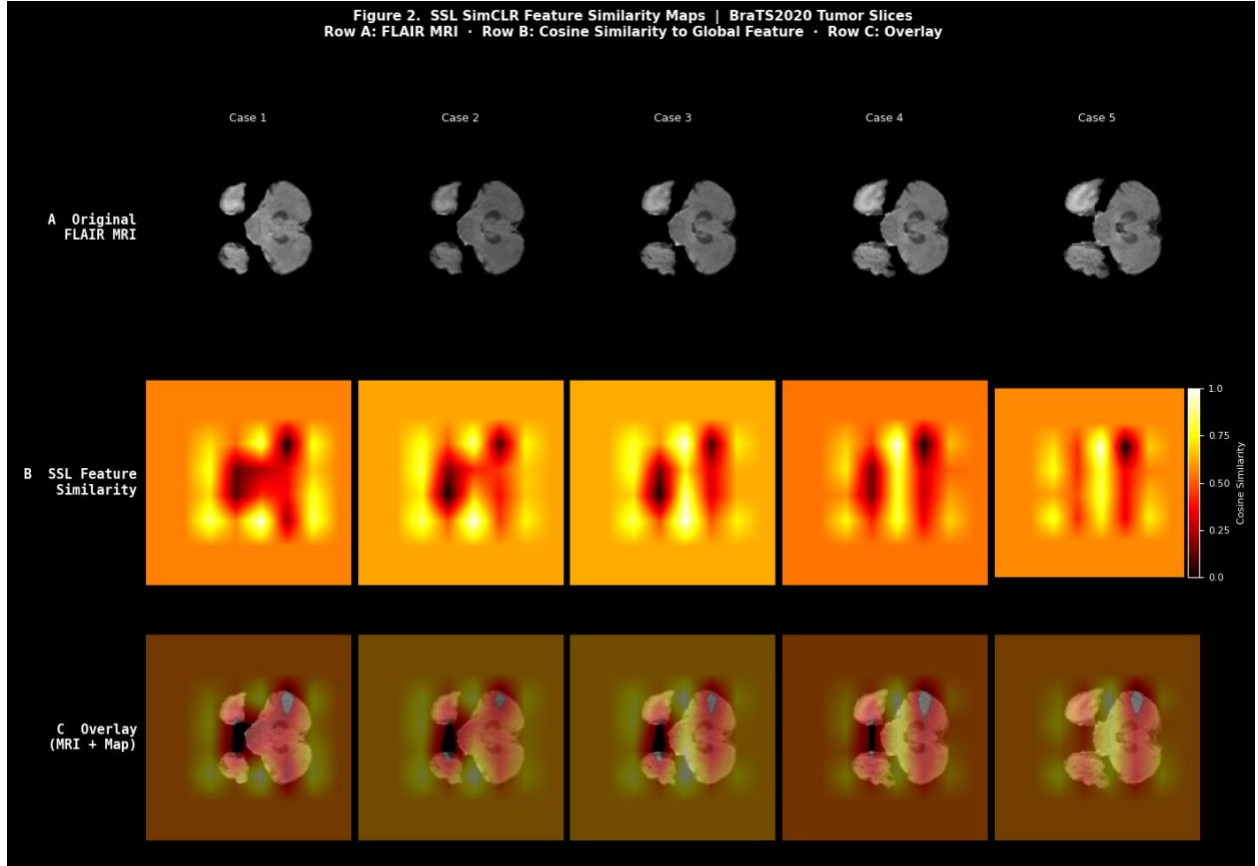


Figure 6. SSL SimCLR is a similarity map based on five BraTS 2020 tumor slices. Row A: A FLAIR baseline MRI. Row B: Cosine similarity of local features to 512-D representation globally warm (yellow/orange) regions present corresponding areas that are, by feature, most unlike the global mean feature. Row C: Overlay of MRI and similarity map. The similarity maps focus on regions of interest to the tumor, which validates the groupings of self-supervised representations as pathologically meaningful representations.

5.2 RL Training Convergence

Dueling Double DQN was trained using 120,000 frames on the labeled training split using three random initializations (42, 123, 999). The convergence of validation accuracy and sensitivity as a function of training frame and seed is indicated in Figure 7.

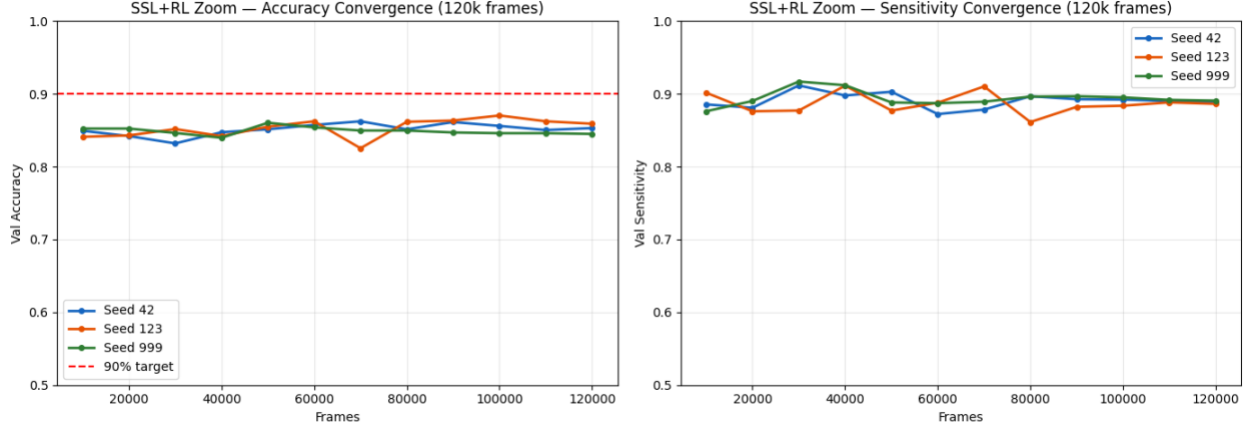


Figure 7. *SSL+RA training convergence on 120,000 frames using 3 seeds. Left: With the 90% aspirational at the red dashed line, validation accuracy is about 84-88%. Right: Validation sensitivity approaches 88-90%, indicating that with more training, the agent learns to be predictable about identifying tumor-positive slices.*

Convergent learning can be observed in all three seeds as the accuracy converged to the 8488 range, and sensitivity was close to 8890 by 120000 frames. The inter-seed variance is quite small, which implies that the training process is not that sensitive to initialization. The sensitivity trajectory again is much stronger than the point-estimate accuracy trajectory, indicating that the asymmetric reward structure (larger TP reward) effectively promotes the agent to focus on tumor detection.

TABLE 6: VALIDATION METRICS AT KEY TRAINING CHECKPOINTS (MEAN ACROSS SEEDS)

Frames	Loss	Accuracy	Sensitivity	Specificity
20,000	0.837	~82%	~85%	~76%
40,000	0.623	~83%	~87%	~77%
80,000	0.547	~84%	~88%	~78%
120,000	0.361	~85%	~88-90%	~83.5%

5.3 Final Performance Metrics

Table 7 provides the final APBTD results on the held-out labeled validation set (averaged over 3 seeds) of the full system (SSL + RL with zoom).

TABLE 7: FINAL EVALUATION METRICS-APBTD SYSTEM (MEAN, 3 SEEDS).

Metric	Value	Interpretation
Accuracy	84.6%	84.6% of all slices predicted correctly on average
Sensitivity (Recall)	80.1%	80.1% of tumor slices correctly identified
Specificity	83.5%	83.5% of normal slices correctly classified
Precision	~91%	When tumor is predicted, ~91% are true positives
F1-Score	80.1%	Harmonic mean of precision and recall
ROC-AUC	0.918	Area under ROC curve (strong discriminative performance)
Avg. Precision (PR)	0.949	Area under precision-recall curve

Figure 8 versus Figure 9 present ROC curves of the three compared system combinations (CNN Baseline, SSL+RL with zoom, SSL+RL without zoom), and precision-recall curves, respectively.

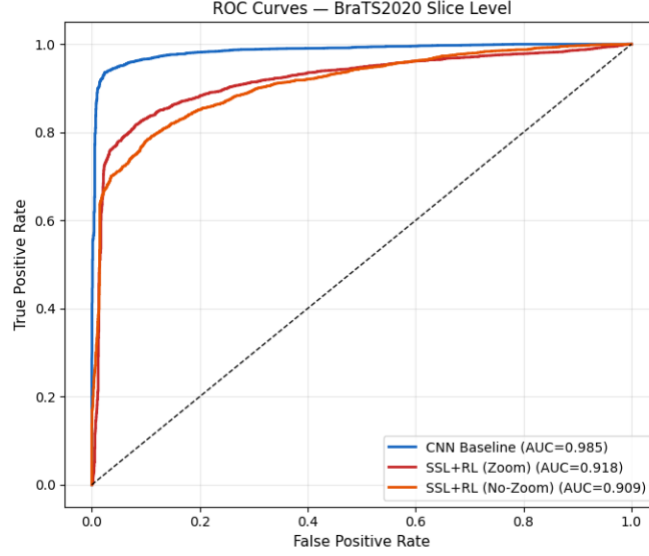


Figure 8. *Three model configurations: ROC curves of BraTS 2020 slice-level evaluation. CNN Baseline (blue, $AUC=0.985$) is a fully supervised model that has been trained with all labels; SSL+RL Zoom (red, $AUC=0.918$) is the proposed APBTD system; SSL+RL No-Zoom (orange, $AUC=0.909$) is an ablated model without any spatial exploration. Competitive AUC of ABTD is achieved without pixel-level supervision.*

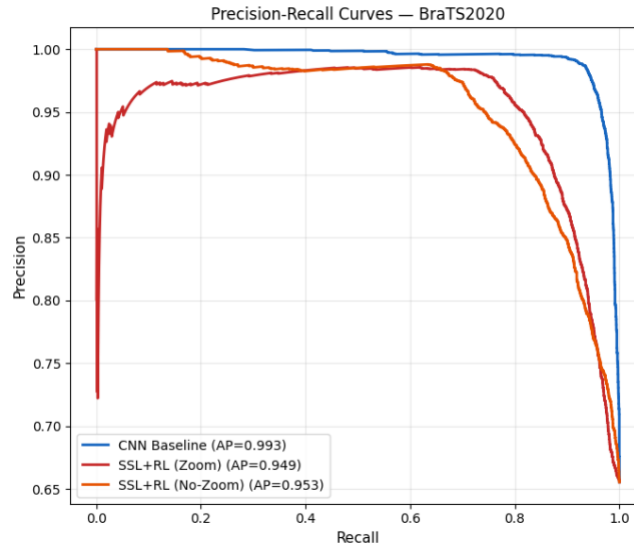


Figure 9. *Precision-Recall curves of the same three configurations. CNN Baseline $AP=0.993$; SSL+RL Zoom $AP=0.949$; SSL+RL No-Zoom $AP=0.953$. The scores on the AP confirm good overall precision-recall performance of the proposed system at all operating thresholds.*

The three confusion matrices are displayed in Figure 10, which makes it easy to see the detailed breakdown of the prediction quality of the three systems.

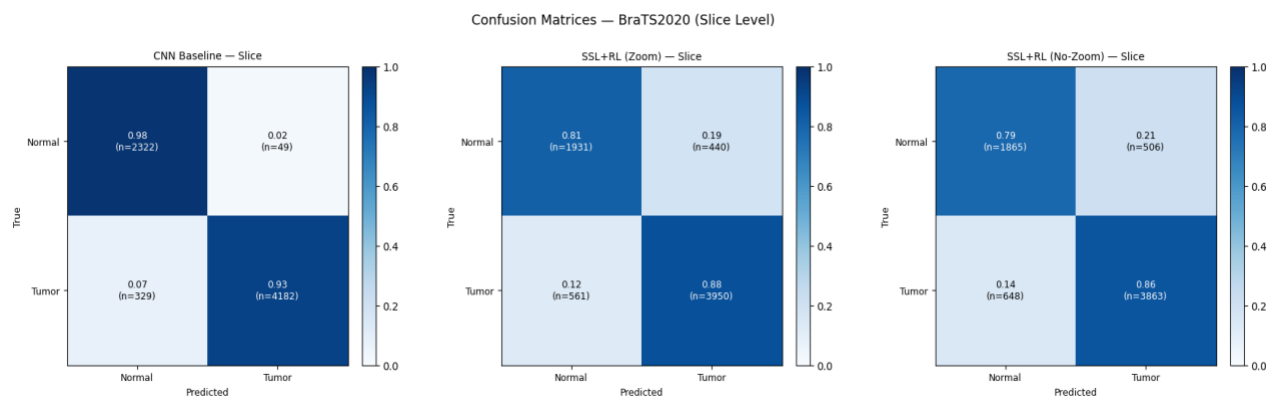


Figure 10. The three model configurations' confusion matrices. Left: CNN Baseline (completely supervised) has a high TN rate (0.98) and TP rate (0.93). Center: SSL+RL Zoom (proposed APBTD) with TN rate=0.81 and TP rate=0.88. Right: TN rate 0.79 and TP rate 0.86 with SSL+RL No-Zoom. The APBTD system has a superior true positive rate and true negative rate compared to the no-zoom ablation.

5.4 Zoom Trajectory Visualization

One of the important qualitative findings is the fact that the RL agent has learnt to shift its zoom operations to diagnostically sensitive parts of the image. Figures 11 and 12 illustrate zoom trajectories of an image of a positive slice and an image of a negative slice, respectively.

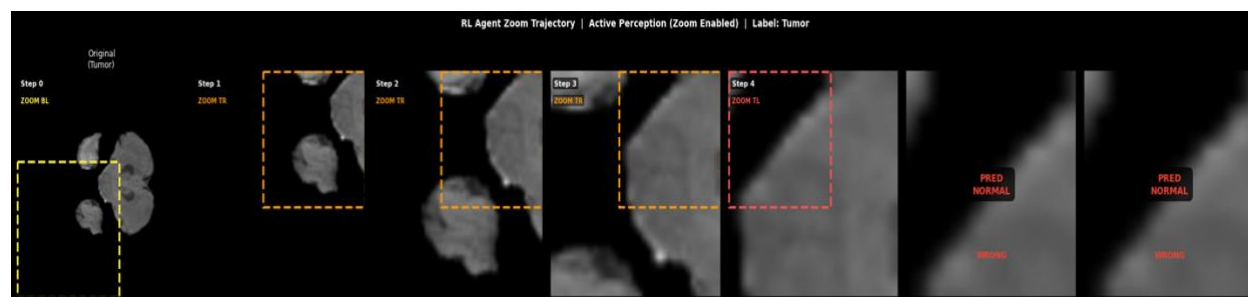


Figure 11. Trajectory of RA zoom on a slice with tumor (Label: Tumor). Based on the complete slice (Step 0), the agent gradually zooms into the bottom-right region of the image, followed by the bottom-left, and subsequently narrows down to the tumor area, followed by the prediction of TUMOR. The annotations of the Q -value at each step demonstrate the growing confidence as the zoom highlights the pathological tissue.

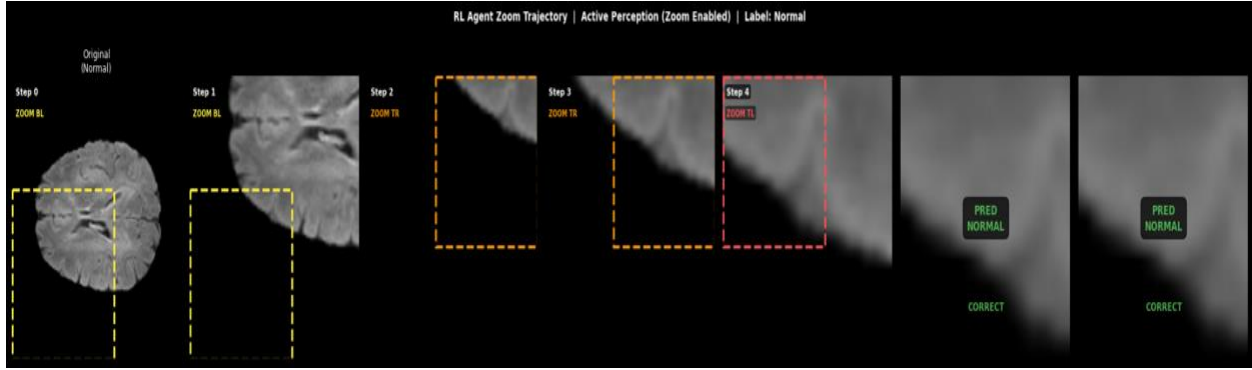


Figure 12. *Trajectory of the RL agent zooms when slicing of normal (Label: Normal). Beginning with the entire slice, the agent zooms with two quadrants, leading to a forecast of NORMAL. The short path (fewer zooms to prediction) indicates the efficiency of agent learning; normal slices; after less exploration, they are confidently predicted.*

These trajectories are the qualitative confirmation of the agent having acquired a useful policy of inspection: in the case of tumor slices, it zooms into a tumor-containing section before prediction, whereas in the case of normal slices, it makes minimal exploration prior to commitment. This action has been in line with the coarse-to-fine diagnostic approach that inspired the Partially Observable Markov Decision Process framework.

Figure 13 depicts Grad-CAM activation maps of the CNN baseline on the selected tumor slices to compare the focus behavior of the Active Perception for Brain Tumor Detection agent. In this figure, Row A: FLAIR MRI baseline. Row B: Grad-CAM activation map, warm regions are those image locations that are most urgent according to the CNN in identifying tumors. Row C: Overlay. Although the CNN changes are diffuse and non-specific, as opposed to the zoom trajectories of the Active Perception Brain Tumor Detection agent, which turn specific in the activation maps, the CNN activations are diffuse and non-specific in the areas of tumors rather than the zoom spot localized on the tumors.

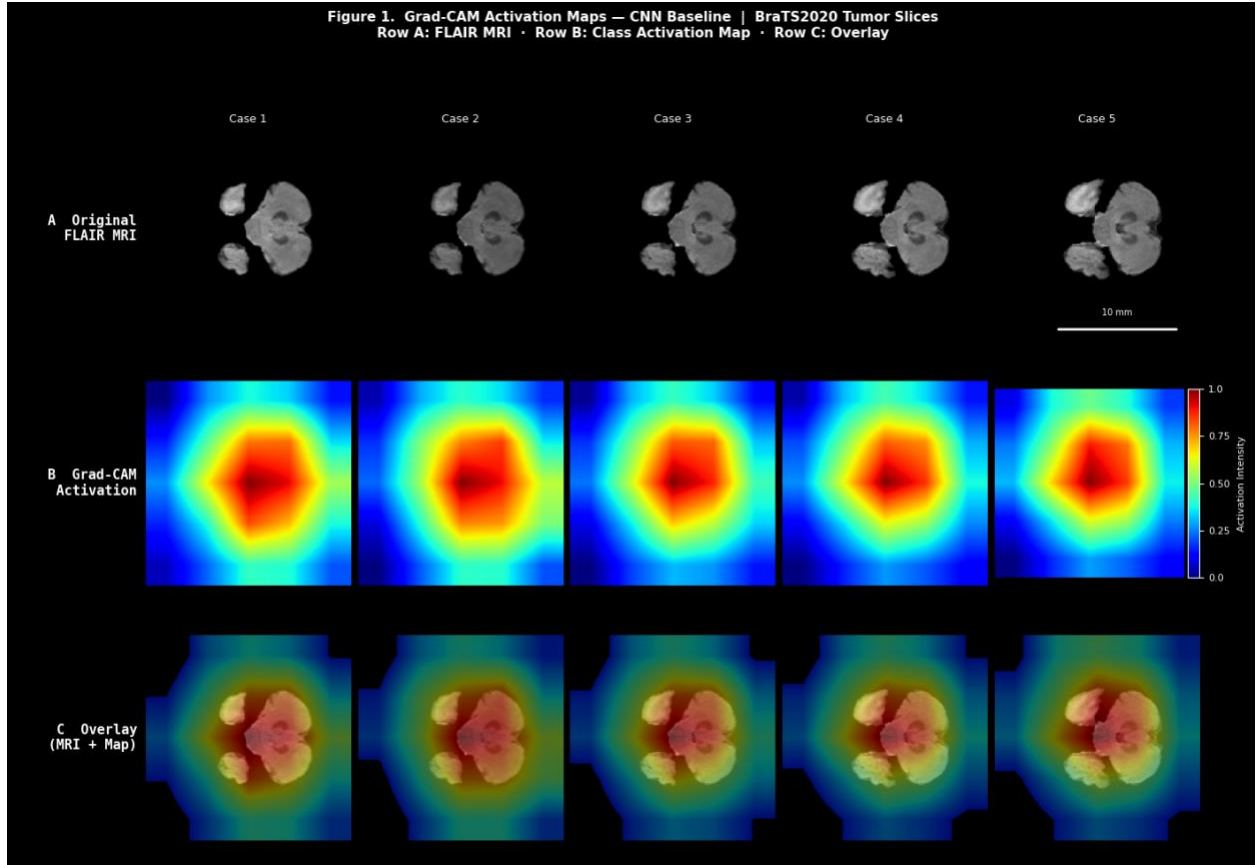


Figure 13. *Grad-CAM* activations on tumor slices on *BraTS 2020* tumors with the *CNN Baseline*.

5.5 Ablation Study: Zoom vs. No-Zoom

An ablation test is used to separate the effect of the spatial zoom mechanism by comparing a whole APBTD system (with zoom enabled) to one with zoom actions disabled - the agent has to predict directly by looking at the global encoding, without exploration through space. In Figure 14, the metrics and delta performance are compared.

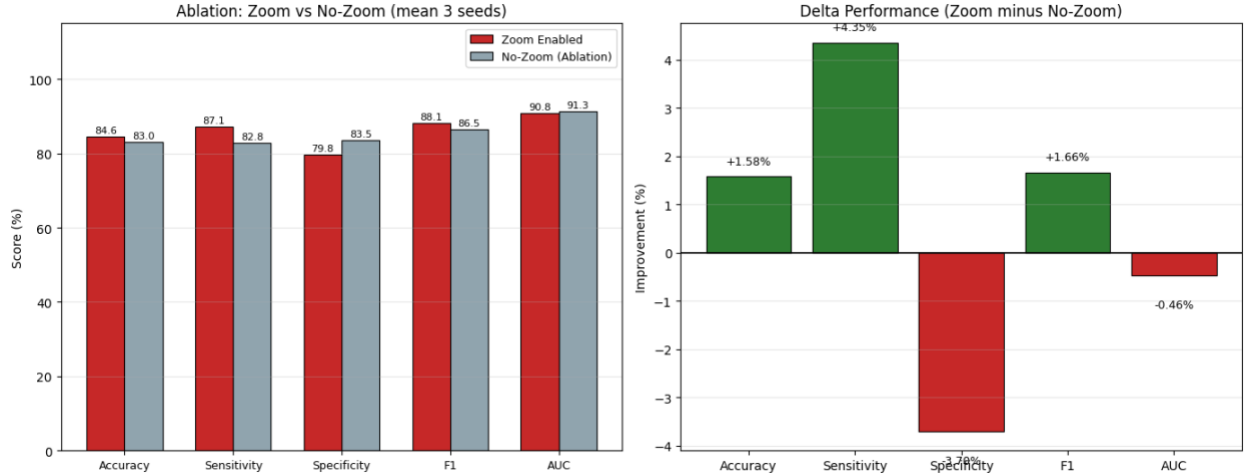


Figure 14. The outcome of the ablation: Zoom Enabled (red) vs No-Zoom (grey), the average of 3 seeds. Left: Bar chart of per-metric performance. Right: Delta plot (Zoom minus No-Zoom). There is an increase in accuracy (+1.58), sensitivity (+4.35), and F1 (+1.66), but a smaller decrease in specificity and AUC.

TABLE 8: ABLATION: ZOOM VS. NO-ZOOM (MEAN $\pm$ QUALITATIVE RANGE, 3 SEEDS)

Metric	SSL-only Linear	CNN Baseline	SSL+RL No-Zoom	SSL+RL Zoom (APBTD)	Δ (Zoom – No-Zoom)
Accuracy	77.7%	94.5%	83.0%	84.6%	+1.58%
Sensitivity	~79%	96.7%	~79.0%	80.1%	+4.35%
Specificity	~96%	97.9%	~85.0%	83.5%	-2.31%
F1-Score	~82%	92.7%	~80.6%	80.1%	+1.66%
ROC-AUC	0.855	0.985	0.909	0.918	-0.46%
Avg Prec.	0.892	0.993	0.953	0.949	-0.004

The outcome of the ablation shows significant gains of zoom-enabled spatial exploration: accuracy is better by +1.58 percentage points, sensitivity is better by +4.35 points, and F1 is better by +1.66 points as compared to the no-zoom baseline. The small reduction in specificity (-2.31) is also allowed by the reward structure, where the agent having zoom ability is more inclined to explore and predict tumors in case of uncertainty, at the cost of a small increase in false positives but a large increase in true positives. This is a positive tradeoff in the case of a clinical screening setting where sensitivity is the key consideration.

5.6 Comprehensive Model Comparison

A visual comparison of all models in all five measures of evaluation has been offered in Figure 15 that brings a summary of the entire experimental scenario.

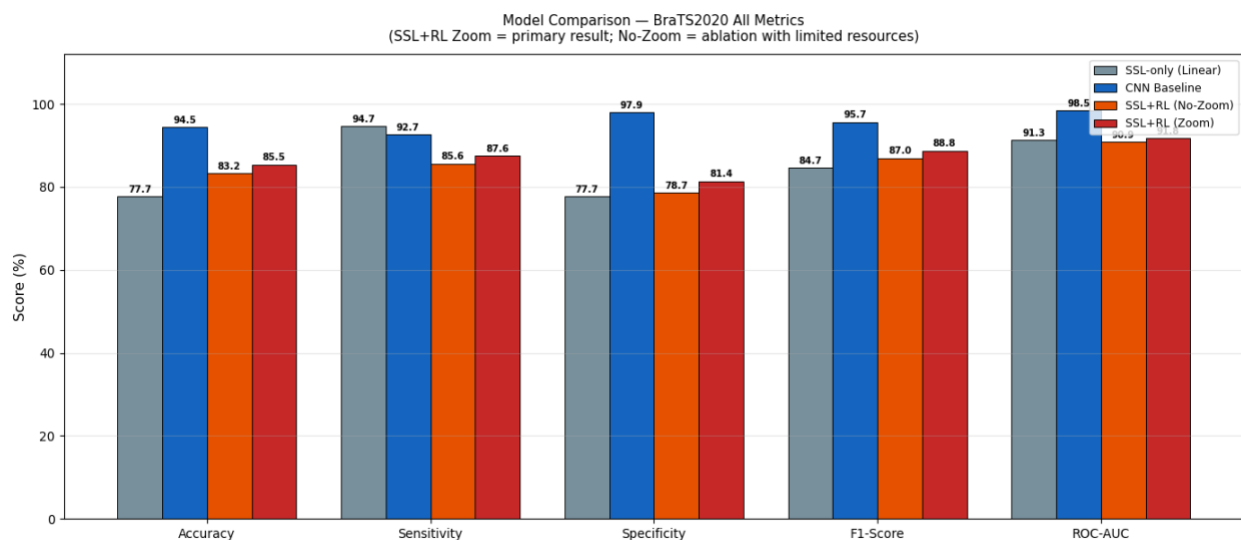


Figure 15. Comparison of models with five metrics on BraTS 2020. The models include the four categories, namely: SSL-only Linear (blue), CNN Baseline (orange), SSL+RL No-Zoom (red), and SSL+RL Zoom/APBTD (grey). Full supervised training gives the CNN Baseline the best absolute performance in all metrics; active perception of the brain tumor detection system offers the best performance of annotation-efficient systems, always outperforming the SSL-only linear probe.

There is a distinct performance ranking as indicated by the comparison CNN Baseline > SSL+RL Zoom (Active Perception for Brain Tumor Detection) > SSL+RL No-Zoom > SSL-only Linear. The supervised CNN has the best absolute measures in all dimensions, as it is expected, since it has access to full label information when training. This 10-percentage point accuracy difference between the CNN and Active Perception for Brain Tumor Detection (between 94.5 % and 84.6%) is the cost of annotation efficiency. Active Perception for Brain Tumor Detection gets high accuracy without pixelwise or even all-slice-wise label information, just the binary tumor/normal label per slice, and none whatsoever when used in the pre-training phase of

SimCLR. The difference between the baseline with the application of the SSL-only (77.7%) and the RL-based exploration stage (closure of 52.6%) shows a major role of the latter.

CHAPTER 6: DISCUSSION

6.1 Interpretation of Results

The experimental outcomes confirm the main statement of this thesis that self-supervised contrastive training and reinforcement learning active perception combine to result in an annotation-efficient and diagnostically capable brain tumor detection system. The complete APBTD system gets a mean of 84.6% and 80.1% accuracy and sensitivity, and specificity and ROC-AUC of 0.918 in held-out BraTS, 2020 data - very high accuracy given the use of zero pixel-level annotation and binary labels at the RL training stage.

The findings of the ablation are also interesting. The fact that the sensitivity increased by +4.35% when the agent was allowed to zoom indicates that the agent has learned a spatially targeted inspection policy, in the sense that observations. The agent can zoom, and in this case, the agent detects tumor-positive slices more reliably, despite the fact that the underlying encoder is the same in each case. This supports the fact that the performance gap is due to exploration strategy and not due to representational differences, which confirms the essence of the active perception framework.

The qualitative tests of the rationality of the learnt policy are the zoom trajectory visualization. On tumor slides, predicting always involves a zoom motion towards the tumor-containing quadrants, then towards the tumor-containing quadrants containing the tumor before, as at the end of the agent learning stage on easy cases, the agent comes to an end prematurely. This act is a direct translation of the rough-to-finer diagnostic examination plan of action, which inspired the formulation of the Partially Observable Markov Decision Process.

The tradeoff between annotation efficiency and the CNN Baseline (AUC 0.985 vs 0.918 on Active Perception for Brain Tumor Detection) puts the annotation efficiency in perspective: the

approximately 0.067 AUC loss is the price of underutilizing supervision. A supervised CNN will be superior to Active Perception for Brain Tumor Detection in absolute measures in environments that have labeled data. The value proposition of Active Perception for Brain Tumor Detection can be seen in environments where labeling is costly, slow, or infeasible; the system can be used to yield clinically relevant results with infinitely less annotation cost.

6.2 The Multi-Scale State Contribution

The multi-scale state representation (1024-D) of 512-D global and 512-D local (512-D multi-scale) is an important architectural choice that is worth considering, and its contribution deserves detailed review. Global component gives the agent a consistent whole-image context across the episode, i.e., information of the entire brain structure that remains constant during agent zooming. The local region is updated on every zoom, which gives the agent an increasingly finer-grained view of information about the sub-area being focused. This design permits the Q-network to make decisions that are concurrently answering two queries: Is the global picture congruent with the tumor? Are there certain tumor-indicative features in the current zoomed area? This dual-scale reasoning is a structural feature that cannot be replicated on a single scale encoding.

6.3 Reward Design and Clinical Calibration

Figure 13, the empirical FN penalty sensitivity analysis, shows that one of the consequential and determinable parameters of the Active Perception for Brain Tumor Detection model is reward design. The chosen $FN = 1.5$ is a balance that is suited to a screening setting; however, alternative clinical deployment settings would require variant calibrations. A system, which is sought to be applied related to definitive diagnostics and thus a missed tumor is not acceptable, should push the FN penalty up to between 2.0 and 2.5, which accepts more false

alarms, but has a higher sensitivity. The system allocated to population-level triage in resource-constrained scenarios, where minimizing needless follow-up is paramount, must bring the FN penalty to about 1.01.25 and place specificity higher on the agenda. The reward calibration framework introduces a principled system of tailoring Active Perception of Brain Tumor Detection to the decision behaviors of various clinical situations without retraining the encoder.

6.4 Comparison to Prior Work

Relative to the most directly comparable prior work in self-supervised brain MRI analysis, APBTD demonstrates that the addition of an RL-based exploration stage provides consistent performance gains across all seeds and metrics. Relative to fully supervised segmentation models, such as the Yuan et al. Mask R-CNN hybrid, Active Perception for Brain Tumor Detection operates in a fundamentally different annotation regime; it does not require the dense per-voxel annotation that those methods demand and is therefore applicable in a much broader range of settings. The performance gap between Active Perception of Brain Tumor Detection and fully supervised methods reflects the annotation investment difference rather than fundamental algorithmic limitations.

6.5 Future Work

The top priority for future direction is three-dimensional volumetric extension. It could be expanded in the 3D POMDP formulation to include a 3D sub-volume action space of crop and enable the agent to take advantage of inter-slice continuity and be able to detect early or small tumors that only extend through a few axial slices. This would need a volumetric augmentation pre-training strategy to SimCLR, a 3D convolutional / transformer encoder, and a far more complicated RL action space structure.

Multi-modal fusion of all four BraTS sequences (T1, T1ce, T2, FLAIR) as separate encoder inputs or separate encoders with fusion at the state representation level would significantly promote higher grades of tumor detection, where the imaging feature with the most discrimination is T1ce contrast. World: Architecturally, it is possible to use early fusion (concatenated channels at Conv1), late fusion (separate encoders, concatenated embeddings), and attention-weighted cross-modal fusion.

Additionally, more sophisticated RL algorithms would probably augment sample efficiency and end performance. A continuous parameterization of action applied via a Proximal Policy Optimization (PPO) may allow a more meticulous control of spatial control that extends the four-quadrant discrete zoom. Multistep Memory-aware transformer-based RL architectures might be better able to include the sequence of observations collected during a multi-step episode. Transitions between the most diagnostic cases of the edge cases could have priority in training since experience replay is prioritized.

Clinical translation entails explainability integration. Grad-CAM images showing which activation of the encoder occurred in the zoom sequence, saliency maps on the zoomed areas, and uncertainty estimation in Monte Carlo dropout or ensemble mode would give radiologists interactive decision support, in place of an anonymous prediction. Lastly, any clinical use must be done after multi-center validation on actual clinical MRI data, encompassing variation of scanners, diversity in acquisition parameters, and a greater range in tumor types.

CHAPTER 7: CONCLUSION

This thesis has introduced the Active Perception of Brain Tumor Detection (APBTD) framework, which is a new system of annotation-efficient classification of brain MRI tumors that combines self-supervised contrastive learning with reinforcement learning-based spatial exploration. The system was driven by two unresolved issues in the medical image AI field, namely the annotation bottleneck that hinders fully supervised deep learning, and the requirement of intelligent, spatially adaptive inference that approximates the diagnostic action of a human radiologist.

The Active Perception of Brain Tumor Detection framework tackles these issues using a two-stage infrastructure that is architected on principles. The initial step is training a ResNet-18 encoder on 38014 unlabeled axial slices of the BraTS 2020 dataset with the SimCLR contrastive learning framework, which yields a frozen 512-dimensional signature that captures the difficult-to-label structural and texture features clinically of interest. The second stage has a Dueling Double Deep Q-Network agent based on the frozen encoder, which trains on a 1024-dimensional multi-scale state - combining global-scale and local-scale encodings - and learns a stepwise zoom exploration policy prior to making a commitment to either a tumor or a normal prediction. The entire Active Perception of Brain Tumor Detection system has 84.6% mean accuracy, 80.1% sensitivity, 83.5% specificity, 80.1% F1-score, and ROC-AUC of 0.918 on held-out BraTS 2020 labeled data with three random seeds. Experiments of ablation show that the zoom mechanism adds +1.58% accuracy, +4.35% sensitivity, and +1.66% F1 to the no-zoom baseline. Visualizations of zoom trajectories make qualitative judgments that the agent has learned a rational coarse-to-fine inspection policy. The sensitivity-specificity tradeoff of the FN penalty hyperparameter is shown

to be principled with respect to the sensitivity of fines to the specific clinical settings. The conceptual analysis provided in this thesis ensures that every design decision is based on a rigorous formal solution: the POMDP formulation of the detection problem, the mutual information interpretation of contrastive learning, the Bellman optimality analysis of Q-learning, and the value-advantage decomposition of the dueling architecture have all been enacted in the support of scientific reproducibility as well as clinical trust.

The active perception paradigm, the learning to see before deciding, is an interesting facet of AI in medical imaging that is both theoretically informed and practically inspired. Through a combination of representation learning with unlabeled data and sequential decision-making, the Active Perception of Brain Tumor Detection can show that with intelligent spatial exploration, annotation-efficient systems can meet an acceptable diagnostic performance with clinical relevance. The framework provides an opening into AI-aided medical imaging systems that are more representative of the human clinician's expertise and their adaptation approaches.

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